Name bhavya rana

Sapid 590034689

Batch 15

Algorithm

1. Read the size of the array, `size`.
2. Declare the variable-length array `arr[size]`.
3. Read all elements of `arr[]`.
4. Read the window size `k`.
5. Pass the address of `arr[]` to `window_maximum()`.
6. If `k <= 0` or `k > size`, display `-1`.
7. Otherwise, use an outer loop from `i = 0` to `size - k` to consider every window.
8. Set `max` to the first element of the current window.
9. Use an inner loop to compare all remaining elements in that window.
10. Update `max` whenever a larger element is found.
11. Display the maximum of each window separated by spaces.

Test Case Table

Test	Array	k	Expected Output
1	1 3 -1 -3 5 3 6 7	3	3 3 5 5 6 7
2	1 2 3 4 5	2	2 3 4 5
3	1 2 3	4	-1
4	5 4 3 2 1	2	5 4 3 2
5	2 8 1 7 3	1	2 8 1 7 3
6	-5 -2 -8 -1	2	-2 -2 -1
7	4 7 2 9	4	9
8	1 2 3	0	-1

Time Complexity: `O((size-k+1) × k)`, which is `O(nk)` in general.

This satisfies the required **nested-loop approach** and uses pointer notation such as `*(arr + i)` to access array elements.

C program

```
#include <stdio.h>

void window_maximum(int *arr, int size, int k)
{
    int i, j, max;

    /* Check if k is invalid */
    if (k <= 0 || k > size)
    {
        printf("-1\n");
        return;
    }

    /* Consider every window of size k */
    for (i = 0; i <= size - k; i++)
    {
        max = *(arr + i);

        /* Find maximum element in current window */
        for (j = i + 1; j < i + k; j++)
        {
            if (*(arr + j) > max)
            {
                max = *(arr + j);
            }
        }

        printf("%d", max);

        /* Print space except after last element */
        if (i < size - k)
        {
            printf(" ");
        }
    }

    printf("\n");
}

int main()
{
    int size, i, k;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    int arr[size];

    printf("Enter %d elements: ", size);

    for (i = 0; i < size; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter k: ");
    scanf("%d", &k);
```

```
 Last login: Sat Oct 24 11:13:16 on ttys000
[bhavya@Bhavyas-MacBook-Air ~ % nano day55.c
[bhavya@Bhavyas-MacBook-Air ~ % clang day55.c
[bhavya@Bhavyas-MacBook-Air ~ % ./day55
 zsh: no such file or directory: ./day55
[bhavya@Bhavyas-MacBook-Air ~ % clang day55.c -o day55
[bhavya@Bhavyas-MacBook-Air ~ % ./day55
 Enter the size of the array: 8
 Enter 8 elements: 1 3 -1 -3 5 3 6 7
 Enter k: 3
 3 3 5 5 6 7
[bhavya@Bhavyas-MacBook-Air ~ % ./day55
 Enter the size of the array: 3
 Enter 3 elements: 1 2 3
 Enter k: 4
 -1
 bhavya@Bhavyas-MacBook-Air ~ %
```

```
    /*
Pass
array
using
its
address
*/
    window_maximum(arr, size, k);

    return 0;
}
```

Execution